

Deriving Adjectives in Heritage Tamil: Stability and Change

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Abstract: Heritage grammars tend to undergo structural change owing to their severely constrained input conditions and /or transfer effects from the L2 (Polinsky, 2018). This study uses adjectives in Tamil (Dravidian) to show a systematic difference between rule-governed, structural aspects of grammar and those that require case-by-case lexical learning. The former remains stable and the latter undergoes change in heritage Tamil. The empirical domain of adjectives in Tamil is novel and particularly informative, as the derived nature of these adjectives helps us identify areas of grammatical stability and those of change, when the context of acquisition diverges from the norm, i.e, heritage grammars. The paper has two major aims: (i) to provide an explanation of adjectives in standard Tamil, and (ii) to inquire into how the derivation of adjectives fares in the context of heritage Tamil. (i) is addressed by showing that adjectives in Tamil are not an independent category in the lexicon, but the derivational component recognises them as a distinct category. We then proceed to question (ii): With respect to heritage Tamil, two domains - one requiring intensive learning, and not requiring learning - are identified. The paper provides novel empirical evidence to demonstrate their stability or variation in heritage grammars.

Keywords: *Derived adjectives, Heritage Language Syntax, Grammatical Change and Variation*

1 Introduction

Heritage Grammars are cases of asymmetric bilingualism where the home language – defined as the L1, or the Heritage language – is different from the dominant language of the society, the L2. These are conditions of acquisition where a child acquires L1 naturalistically at home while simultaneously being bilingual in the L2. In terms of the Primary Linguistic Data (PLD) available to the bilingual child, the abundance of L2 PLD – as the lingua franca, medium of

education, media content, etc – contrasts with limited nature of L1 PLD which is restricted only to the home domain with family members. Reduced PLD coupled with potential cross-linguistic influence from the prevalence of L2 makes the grammar of L1 prone to change and variation. A combination of these factors make heritage languages diverge from their respective baseline varieties in certain domains of the grammar (Polinsky 2018, Rothman 2009).

This paper considers the case of Tamil (Dravidian) as a heritage language in New Delhi, India. The focus is primarily on the domain of adjectives, and the role played by reduced PLD in the home language. The first objective of the paper is to provide a sketch of adjectives in Tamil. Based on the inferences drawn from a theoretical understanding of adjectives, Section 4.1 formulates hypotheses and predictions for how we can expect adjectives to fare in Heritage Tamil.

2 Rationale behind the study

It has been widely recorded that heritage grammars undergo change and variation from their baseline owing to factors like restricted PLD and cross-linguistic effects from the dominant societal language. However, not all aspects of grammar are affected equally by these factors. Some grammatical phenomena tend to be more resistant to change than others (Polinsky 2015, Unsworth et al. 2014). One indicator of a grammatical phenomenon being prone to change in a heritage bilingual context is that it involves intensive learning on the part of the language acquirer. Lexical items, idiosyncrasies, case allomorphs, etc., are areas where heritage grammars often display divergence from the baseline (Polinsky 2018). Such a trend can be explained as a function of reduced PLD; if a grammatical phenomenon has to be learned on a case-by-case basis, with knowledge of exceptions and idiosyncrasies, one requires sufficient data for the learning to take place effectively. However, heritage language situations generally do not offer such an environment. As a result, these domains are not fully developed.

On the other hand, we have regular, rule-governed domains which tend to be unaffected in these settings. The rationale is that grammatical phenomena which are part of core syntax come about as part of the UG component. These do not need to be learned individually. Thus, these do not suffer for the lack of PLD. In several studies in the field, (Lohndal et al. 2019, Polinsky 2018, Unsworth et al. 2014) it is empirically attested that core syntax phenomena develop unhindered even in such asymmetrically bilingual contexts of acquisition.

The current study is an attempt to contribute to the bifurcation between stable and vulnerable domains in heritage languages. The empirical ground of this study is formed by Tamil as a heritage language. Tamil is a Dravidian language primarily spoken in Tamil Nadu, India. In this study, I focus on Tamil acquired naturalistically in New Delhi, the capital city of India. The lingua franca of New Delhi is Hindi-Urdu, an Indo-Aryan language. Large-scale but incremental migration from Tamil Nadu to New Delhi during various times in the 20th century led to several speakers of Tamil taking up permanent residence in New Delhi. The next generation, often born in New Delhi, acquired Tamil as the first language and Hindi-Urdu as a simultaneous second-language. Tamil is therefore restricted only to the home domain, and Hindi-Urdu finds prevalence in all other spheres of their lives; the immediate neighbourhood has Hindi-Urdu speakers, it is taught as the second language in many schools, it is

also the language in which they mostly consume mass media content, and the language used for formal as well as informal communication in the city. These second/third generation immigrants are defined as heritage speakers of Tamil in New Delhi.

The grammatical phenomenon that will be investigated in heritage Tamil is the domain of adjectives. The reason for selecting adjectives as the focus of the study are two-fold: (i) As we shall see in the forthcoming section, adjectives are a complex category in Tamil. They are formed syntactically using specific derivational processes - thus making them an insightful object of study in a grammar that could undergo change. (ii) Within adjectives there are two sub components: the formation of these adjectives and executing agreement relations wherever applicable. The former is a domain requiring intensive learning, whereas the latter is part of core syntax, thus enabling us to monitor how each of them fares in a grammar acquired under input-related constraints.

Following from the ideas laid out so far, the overall objective of this paper is to identify which grammatical domains are more susceptible to change and which ones display relative stability. To that end, this paper investigates one domain which requires learning and one which does not. The advantage of comparing such a dyad in the same language is that it allows to zoom in on the nature of certain grammatical phenomena that could be inherently more prone to change than others. It effectively makes it possible to keep all other factors constant and compare only one aspect of the grammatical phenomena in question.

3 Structural representation of adjectives in Tamil

This section presents an overview of the structural representation of adjectives in Tamil. Adjectives in all Dravidian languages, including Tamil, have been the topic of considerable debate in the literature. The main question is whether or not they exist as an independent category in the language. The first part of this section unpacks the existing accounts of adjectives in Tamil. Subsequently, I place the current proposal that adjectives do exist in Tamil; they are derived syntactically. Following this, the exact underlying representations of attributive and predicative adjectives are explained.

3.1 The on-going debate

Existing accounts of the notion of adjectival modifiers in Tamil and other Dravidian languages adopt the position that these languages simply lack the category of adjectives. The claim is a gradient one, with varying levels of strength. The strongest version of the claim is that Tamil lacks adjectives altogether (Menon 2013, 2014). A more flexible line of thinking draws on the lexicon-syntax divide (Ramchand 2024), and propose that while the Tamil lexicon may not represent adjectives as an independent category, the syntactic component may be able to identify adjectives as distinct from other categories (Amritavalli 2019, Jayaseelan & Amritavalli 2017). Yet another view, coming from a diachronic empirical perspective, lays out that there are certain modifiers in Tamil that cannot be categorized as nominal, verbal or any other categories, hence forcing us to reconcile with the idea that adjectival modifiers exist in a category of their own (Krishnamurti 2003).

In this section, I present empirical evidence – in the form of distributional restrictions – which clearly demonstrate that the grammar recognizes adjectives as a separate class of lexical items. The chief inference we can draw from these diagnostic tests is that there must be systematic representations underlying adjectives even in a language such as Tamil, where adjectives do not present as an independently defined class in the lexicon. Since the focus of this section on the positions of occurrence of the adjectives, it does not expand on their internal structure. For the purposes of the diagnostic tests, the adjectives (1)-(4) are glossed according to their English counterparts. The exact nature of their structural make-up is presented in Section 3.2.

Distributional restrictions can demonstrate that adjectives are treated as a separate class in the grammar. Firstly, the derived adjectives in Tamil follow certain paths that are well-established as defining properties of adjectives.

(i) Fixed position in the DP: Adjectives have a fixed position in the DP. Languages vary in terms of whether the adjective precedes (A - N) or follows (N - A) the noun. In (1) we can see that the modifiers in Tamil have a fixed pre-nominal position (1a). It is ungrammatical to move it post-nominally (1b).

- | | | |
|-----|--|--|
| (1) | a. anda nalla paiyan
that good boy
'that good boy' | b. *anda paiyan nalla
that boy good |
|-----|--|--|

(ii) Co-occurrence with other modifiers: Another key hallmark of adjectives is that cross-linguistically, they can co-occur with other modifiers such as numerals and quantifiers, iteratively modifying the head noun. The modifiers in Tamil too follow this pattern: (2)

- | | | |
|-----|--|--|
| (2) | a. anda naalu nalla pasanga
that four good boys
'those four good boys' | b. neraya uyaramaana pasanga
many tall boys
'many tall boys' |
|-----|--|--|

(iii) Modification by Intensifiers: Adjectives are the only part of speech that can be modified by an Intensifier - 'very'. This property does not extend to nominal and verbal entities. (3) shows in Tamil too, intensifiers can be applied to derived adjectives (3a) but not to nominals (3b), (3c).

- | | | |
|-----|--|---|
| (3) | a. anda romba nalla paiyan
that INTF good boy
'that very good boy' | c. *anda romba uyaram
that INTF height
Intended: 'that very height' |
| | b. *anda romba paiyan
that INTF boy
Intended: 'that very boy' | |

(iv) Degrees of Comparison: Adjectives are typically gradable: they can be expressed in comparative and superlative terms. The modifiers in Tamil (4) too occur in these forms.

(4)

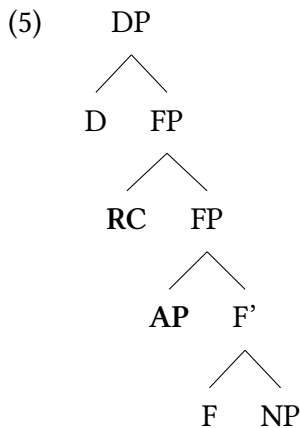
- a. john peter-a vida nalla-van
 John Peter-ACC than good-MSG
 ‘John is better than Peter.’
- b. mary ellarayum vida uyaramaa
 iruk-aa
 Mary everyone.ACC than tall be-FSG
 ‘Mary is taller than everyone.’

With these empirical diagnostics, it becomes evident that adjectives, while not being defined in the lexical, exist as an independent class in the derivational component of Tamil.

3.2 The current proposal

The position adopted in this paper is that adjectives do exist as a distinct category in syntax in Tamil. The next aspect of Tamil adjectives is their structural representation. Based on the morphological shape of the adjectives presented in Section 2, it is possible to infer that adjectives in Tamil are not unary items. They are composite, and therefore must have been derived using designated derivational steps. In this section, we shall uncover these derivational processes and how they lead up to the final form of the adjectives.

Cinque (2010) and Alexiadou (2014) present a unified account of the syntax of adjectives where it is posited that there are two ways in which adjectives are represented across the world’s languages. Adjectives are either merged as phrasal specifiers of dedicated functional projections (AP), or they are introduced into the derivation as predicates of (reduced) relative clauses. Both the paths are illustrated in (5).

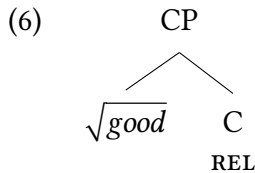


I interpret Cinque’s proposal to mean that the choice between the options presented in (5) is subject to cross-linguistic variation. Languages where adjectives are independent lexical items, already classified as A in the lexicon, can project an AP. In these languages, APs are then directly merged into the structure. Germanic and Romance languages fall into this category. On the other hand, when languages do not define adjectives in the lexicon, as is the case with Tamil, APs are not projected. These languages then require additional mechanisms to realize adjectives in the structure. In the absence of dedicated APs, relative clauses step in to introduce adjectival modifiers.

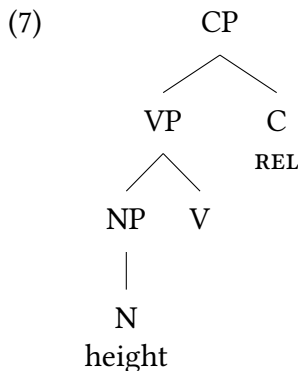
The review of Tamil empirical facts provided in Section 2 makes it evident that there are two types of adjectives. In terms of their structural composition, the implication then is that

they must be formed out of two different kinds of elements. The role of relativisers is crucial to the formation of both these types of adjectives in Tamil, dubbed Type 1 and Type 2 for the moment.

The strategy for forming Type 1 adjectives in Tamil is to apply relativisers directly to property-denoting concepts. Property denoting concepts such as $\sqrt{good}$, $\sqrt{bad}$, $\sqrt{small}$, etc. exist in Tamil. They have phonological forms, but these cannot form morphological words on their own. They necessarily require additional steps to be executed before they can be used in a derivation. (6) illustrates this process. Merging the concept $\sqrt{good}$ with the relativiser leads to the formation of the adjective ‘good’ in Tamil.



The second strategy, Type 2, is adopted to form adjectives out of lexical nouns. Nouns such as ‘height’, ‘happiness’, ‘patience’, etc. can form adjectives with the help of additional structure. In (7), the ‘height’ is merged with a verbal element. The entire complex is merged to a relativizer, with the entire composite unit being interpreted as ‘which is tall’ or simply ‘tall’.



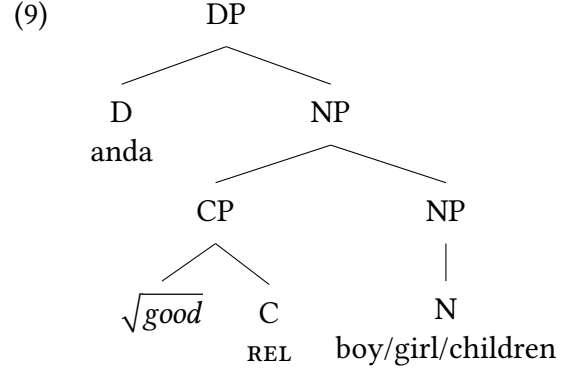
With these skeletal representations in place, we can proceed to look into the particular structural properties of these adjectives. As is true of adjectives cross-linguistically, there are attributive and predicative instantiations of the adjectives in Tamil. These positions accommodate adjectives derived using both strategies listed above.

3.2.1 Attributive adjectives

Attributive adjectives are defined positionally. They occur inside the nominal domain, and are present as additional modifiers of the head noun (8) and (10). In Tamil, adjectives do not participate in agreement relations while in the attributive position. They remain invariant for the phi features of the noun being modified by them. These characteristics are common across adjectives formed via both strategies of adjective formation.

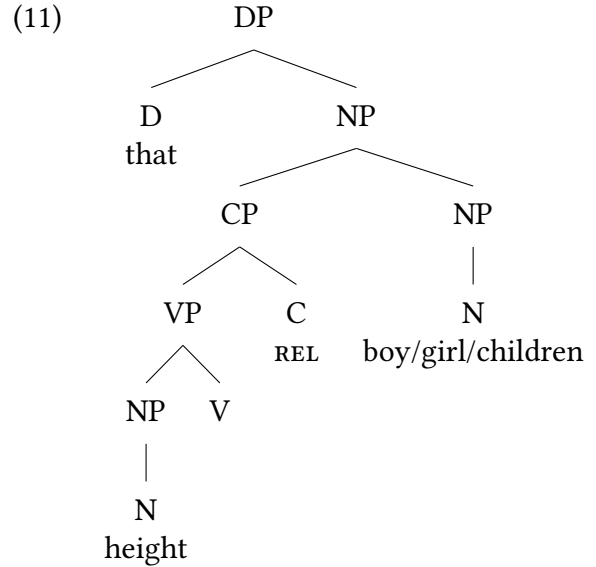
For expository purposes, attributive adjectives are organized by the strategy they were formed by. (8) are the adjectives formed by the strategy in (9), where a property concept is directly merged with the relativizer to form the adjective. The property concepts are not capable of forming morphological words by themselves. In order to be realized, they need the support of other derivational morphemes - a task performed by the relativizer in this instance.

- (8) a. anda nall-a paiyan
that $\sqrt{\text{good}}$ -REL boy
'that good boy'
- b. anda nall-a poNNu
that $\sqrt{\text{good}}$ -REL girl
'that good girl'
- c. anda nall-a koLandainga
that $\sqrt{\text{good}}$ -REL children
'those good children'



Adjectives which originate as lexical nouns are presented in (10). Their structural representation is given in (11), where the noun is first affixed to a verbal element¹, and the entire complex is subsequently verbalized.

- (10) a. anda uyaram-aa-na paiyan
that height-V-REL boy
'that tall boy'
- b. anda uyaram-aa-na poNNu
that height-V-REL girl
'that tall girl'
- c. anda uyaram-aa-na koLandainga
that height-V-REL children
'those tall children'



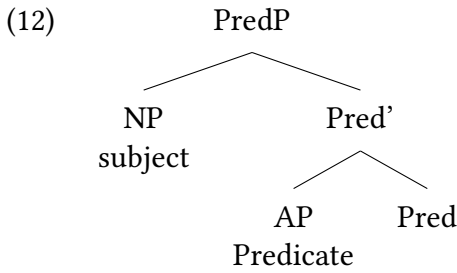
It is worthwhile to note that there are no agreement relations in the attributive domain. Examples (8) and (10) illustrate that the shape of the adjective remains invariant for the changing phi features of the head noun. The absence of agreement in this domain can be explained as a result of there not being any functional head in this domain. Conventionally,

¹There is independent empirical evidence to show that the lexical content of the verb is 'happen', resulting in the structure bring height-happen-REL

agreement is executed by functional heads acting as probes. The attributive domain lacks functional heads, a consequence of which is the absence of agreement in this domain.

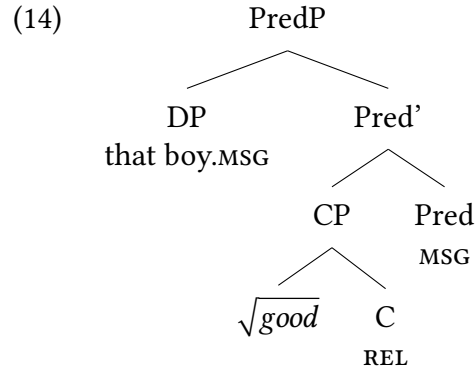
3.2.2 Predicative adjectives

Having laid out the properties of attributive adjectives, we now proceed to the domain of adjectives occurring as predicates. In this structural configuration, the adjective takes the noun as its argument. The structural relation between an argument and predicate is executed as a by a dedicated functional projection, such as PredP for instance. Baker (2008, 2011) proposes that the function of Pred is to establish the structural relation of predication between an adjective and a noun. Structurally, Pred takes an adjective as its complement, and when a noun occupies the specifier of PredP, it is recognised as an argument of the adjectival predicate, represented in (12).



The structure presented in (12) is the one underlying predicative adjectives in Tamil. At the outset, both strategies of adjective formation are found in the predicative domain as well. (13) exemplifies predicative adjectives formed by directly relativising the root. Structurally, these adjectives are CP complements of the functional head Pred (14).

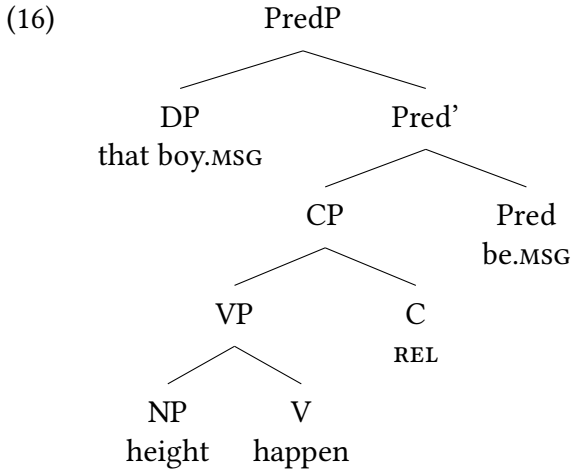
- (13)
- a. anda paiyan nall-a-van
that boy.MSG $\sqrt{\text{good}}$ -REL-MSG
'That boy is good.'
 - b. anda poNNu nall-a-va
that girl.FSG $\sqrt{\text{good}}$ -REL-FSG
'That girl is good.'
 - c. anda koLandainga nall-a-vanga
that children.PL $\sqrt{\text{good}}$ -REL-PL
'Those children are good.'



Adjectives formed through the second strategy are also present in the predicative domain (15). Structurally, these originate as nouns, and are applied to a verbal element before the application of a relative clause (16).

- (15)
- a. anda paiyan uyaram-aa(-ga) iruk-aan
that boy.MSG height-v(-REL) be-MSG
'That boy is tall.'

- b. anda poNNu uyaram-aa(-ga) iruk-aa
 that girl.FSG height-v(-REL) be-FSG
 ‘That girl is tall.’
- c. anda koLandainga uyaram-aa(-ga) iruk-aanga
 that children.PL height-v(-REL) be-PL
 ‘Those children are tall.’



A relevant point about predicative adjectives formed by both strategies (13) and (15) is that there is agreement between the subject and the predicate, reflected on the shape of the predicative adjective. The functional head responsible for agreement in these constructions is Pred (Baker 2011). Pred probes for a DP within its c-command domain to agree with, and subsequently agrees with the subject DP.

3.2.3 Interim Summary

From the discussion in the preceding section, two major characteristics of adjectives emerge very clearly. Firstly, adjectives are not an inherently defined lexical category in Tamil. They are derived syntactically using two strategies. The two strategies have strict domains of application; roots have to be directly relativised, and nouns have to be verbalised before relativising in order to form adjectives. Secondly, once the adjectives are formed, the attributive and predicative adjectives differ in terms of agreement relations. Attributive adjectives do not agree with the noun they modify, whereas predicative adjectives agree with the phi features of the subject DP. With these properties in place, we can now investigate how they fare when Tamil is acquired in a constrained context - as a heritage language.

4 The derivation of adjectives in heritage Tamil

Studies on heritage language syntax are informative, as they allow us to narrow down on a context of acquisition where the PLD available to the language acquirer is more restricted than the norm. The overarching goal of this research area is to identify domains of grammar which are impervious to the effects of the divergent nature of the acquisition context, and

those which develop with some divergence as a result of the acquisition norms. In the current study, we focus on adjectives in Tamil, and investigate the repercussions of the context of acquisition on this domain.

4.1 Hypotheses and Predictions

In this paper, we have laid out two different aspects to Tamil adjectives: their ontology (application of appropriate strategies) and the agreement relations they participate in. The two of them differ fundamentally in terms of how they come about in a grammar. Application of appropriate adjective formation strategies is a domain that requires a considerable amount of learning on the part of the acquirer. In other words, a person acquiring the grammar of Tamil has to be cognizant of the fact that there are two ways of forming adjectives. It has to be further learnt that the two strategies apply to specific, mutually exclusive domains. Such learning is crucially contingent upon the availability of sufficient PLD in the linguistic environment of the acquirer. Studies on heritage language syntax have generalized that (Lohndal 2021, Lohndal et al. 2019, Polinsky 2018) grammatical domains which require lexical learning tend to experience divergence in heritage grammars.

The second aspect of Tamil adjectives explored here is the agreement-based difference. Attributive adjectives do not exhibit agreement owing to the absence of probes in the nominal domain, whereas the presence of pred in the predicative domain facilitates agreement there. This property is a part of core syntax, in a theory where agreement is taken to be a consequence of probe-goal relationships (Chomsky 2000, 2001). As such, it is a part of the regular, rule-governed domain of grammar, and does not require explicit learning. Thus, despite the constrained contexts of acquisition, these domains develop unimpeded. Thus, the predictions for adjectives in heritage Tamil are as follows:

- There will be a considerable overlap in the application of the two strategies of adjective formation.
- Agreement-based distinctions will remain in heritage Tamil. Attributive and predicative adjectives will continue to be distinguished based on agreement.

4.2 Methods and materials

In order to test out the predictions listed above, I conducted a picture-based elicitation task with 24 speakers (mean age=21.4) of heritage Tamil. In order to elicit the production of adjectives, the speakers were shown images of people and a few property-denoting words in English. The prompt given to them was to form sentences by describing the image using the property.

An example of the stimuli used for attributive adjectives is presented in Fig.1 . The participants were asked to form sentences with the pictures in front of them as the subjects of the sentence.

This prompt was expected to elicit utterances such as (17).



Figure 1 Prompt for Attributive Adjectives

- (17) a. anda nall-a paiyan paadar-aan
 that $\sqrt{good}$ -REL boy sing.PRS-MSG
 ‘That good boy is singing.’
 b. anda uyaram-aa-na paiyan toongar-aan
 that height-v-REL boy sleep.PRS-MSG
 ‘That tall boy is sleeping.’

For the predicative adjectives, the stimuli shown were of the kind in Fig.2. In this case, the participants were asked to produce sentences with adjectival predicates using the property-denoting words shown to them.

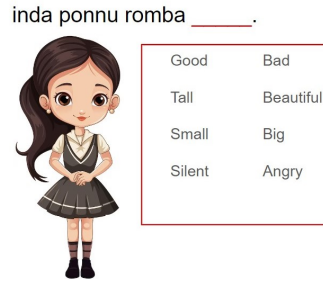


Figure 2 Prompt for Predicative Adjectives

The expected utterances for these prompts are given in (18).

- (18) a. anda poNNu nall-a-va
 that girl $\sqrt{good}$ -RELFSG
 ‘That girl is good.’
 b. anda poNNu uyaram-aa-ga iruka
 that girl height-v-REL be.PRS.FSG
 ‘That girl is tall.’

4.3 Results

The results of the elicited production task confirm prediction 1 mostly. Of the 284 tokens of predicative adjectives collected in the task, around 65% of the utterances were target-consistent. Interestingly, 35% of the utterances showed an instance of the speakers using the two strategies of adjective formation interchangeably. Essentially, we report instances such as (19) and (20) in the data. (19) shows an instance where a type 2 adjective (formed from nouns) is formed using the strategy of type 1. The noun ‘height’ is verbalised. But, instead of using an overt copula, agreement morphemes are directly affixed onto the relativiser.

- (19) inda paiyan uyaram-aa-na-van
 this boy height-V-REL-MSG
 Intended: ‘This boy is tall.’

(20) shows further innovation on the part of the speakers. This is an instance where both the strategies are applied simultaneously to an adjective. A root is first relativised, and the agreement morphemes are affixed to it, as expected. However, the derivation is continued; the adjective is further verbalised and there is an overt copula in the sentence - thereby creating a structure more complex than the expected utterance.

- (20) inda poNNu nall-a-vaL-aa irukaa
 this girl $\sqrt{\text{good}}$ -REL-FSG-V be.FSG
 Intended: ‘This girl is good.’

The second prediction was regarding the structural differences between attributive and predicative adjectives. The primary distinction between the two can be explained in terms of agreement relations. The attributive domain lacks functional heads, and consequently has no agreement between the adjective and the noun it modifies. In contrast, the predicative domain is marked by the functional head Pred, which facilitates agreement between the subject and the adjectival predicate. For heritage Tamil it was predicted that the distinction would be maintained, as it is one that is part of core syntax.

This prediction was attested too. All the attributive adjectives produced by the speakers were devoid of agreement relations, and all but two instances of predicative adjectives occurred with appropriate agreement morphemes.

4.4 Discussion

The results of the present study reveal two fundamental aspects about how heritage grammars are likely to undergo change. To elaborate, we identified two distinct sub-domains within adjectives in Tamil: the formation of the derived adjective, and agreement relations in predicative adjectives. The former is a domain that requires considerable lexical learning on the part of the language acquirer. Heritage Tamil, with its restricted access to PLD, is expected to be unable to effectively facilitate such learning. As a consequence, we observe several instances of divergence from the baseline. Specifically, we observe that the distinction between the application of the two strategies of adjective formation is not maintained strictly in heritage Tamil. The second aspect addressed was agreement. Attributive

adjectives in Tamil do not participate in agreement, whereas predicative adjectives do. This disparity between the two has been explained with the absence and the presence of functional heads in the respective domains. This particular property of the grammar does not require any learning; it is a core syntactic property of grammars under standard assumptions that agreement is carried out by functional heads. Thus, the lack of sufficient linguistic experience does not have any impact in this domain.

The contrasting nature of the heritage grammar in these two domains is particularly informative. As outlined earlier, these grammars develop under constrained contexts of acquisition; the linguistic input received by the language acquiring child is considerably lesser than the norm in a typical context of acquisition. With studies such as these, we are able to locate domains of grammar that emerge resilient despite these factors, and those that are susceptible to change owing to these factors. Syntax then emerges as a resilient domain.

5 Conclusion

The current study is centered around the idea that, when the context of acquisition diverges from the norm, some parts of grammar develop in a stable manner while others undergo change. The empirical demonstration for this idea is provided by Tamil acquired as a heritage language in New Delhi. Using derivationally formed adjectives in heritage Tamil, we can observe the contrast between core, structural aspects of adjectives and those that require case-by-case lexical learning. The former is unaffected, whereas the latter undergoes change. To conclude, this study explains how adjectives are derived in Tamil, and identifies grammatical domains that are susceptible or resistant to change when the grammar develops in an atypical context of acquisition.

Abbreviations

INTF = Intensifier, ACC = accusative, F = feminine, M = masculine, PL = plural, PLD = Primary Linguistic Data, PRS = present, REL = relative, SG = singular.

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